H.T No: | | | | | | | | | | **R18** Course Code: A30005

CMR COLLEGE OF ENGINEERING & TECHNOLOGY

(UGC AUTONOMOUS)

B.Tech II Semester Supplementary Examinations July/August-2024

Course Name: ODEs and Multivariable Calculus

(Common for all Branches)

Date: 15.07.2024 FN **Time: 3 hours** **Max.Marks: 70**

(Note: Assume suitable data if necessary)

PART-A

Answer all TEN questions (Compulsory)

Each question carries TWO marks. **10x2=20M**

1. Solve $y = a\sqrt{(1+p^2)}$ 2 M
2. State law of natural growth or decay. 2 M
3. Solve $(D^2+4)y = sin2x$ 2 M
4. Find the wronskian of two functions $sinx$ and $cosx$ 2 M
5. Evaluate $\int_1^2 \int_0^x y^2\, dxdy$. 2 M
6. Sketch the region of integration for $\int_0^a \int_0^{\sqrt{a^2-x^2}} f(x,y)\, dxdy$. 2 M
7. If $\bar{a}$ is constant vector then find the grad $(\boldsymbol{a}\cdot\bar{r})$ 2 M
8. Find the greatest value of the directional derivative of the function $\emptyset = x^2yz^3$ at (2,1,-1) 2 M
9. State Gauss divergence theorem. 2 M
10. Evaluate $\int_c F\,.dr$ where $\bar{F} = x^2y^2\boldsymbol{i} + y\boldsymbol{j}$ and the curve $y^2 = x$ in the xy-plane from (0, 0) to (4, 4) 2 M

PART-B

Answer the following. Each question carries TEN Marks. **5x10=50M**

11.A). i) Solve $(x+2y^3)\frac{dy}{dx} = y$. 5M

ii) Solve $p^2 - p - 6 = 0$ where p=dy/dx 5M

OR

11. B). Uranium disintegrates at a rate proportional to the amount present at any instant. If M_1 *and* M_2 are grams of uranium that are present at times T_1 *and* T_2 respectively, find the half -life of uranium. 10M

12. A). Solve $\left(x^3\frac{d^3y}{dx^3} + 3x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + 8y\right) = 65\,cos(logx)$ 10M

OR

12. B). Solve $(D^2+1)y = secx$ by the method of variation of parameters 10M

13. A). By changing into polar coordinates, evaluate $\iint \frac{x^2y^2}{x^2+y^2}\, dxdy$ over the annular region between the circles $x^2+y^2 = a^2$ and $x^2+y^2 = b^2$ 10M

OR

13. B). Change the order of integration in $\int_0^a \int_{\frac{x^2}{a}}^{2a-x} x\,y^2\, dxdy$ and hence evaluate the same 10M

14. A). i) Find the value of a and b so that the surfaces $ax^2 - byz = (a+2)x$ and $4x^2y + z^3 = 4$ may intersect orthogonally at the point $(1,-1,2)$. 5M

ii) Find curlF at the point (1,2,3) given that $\boldsymbol{F}$ = grad $(x^3y + y^3z + z^3x - x^2y^2z^2)$. 5M

(P.T.O..)

OR

14. B). Find constants a, b, c so that the vectors $\bar{A} = (x + 2y + az)\bar{\imath} + (bx - 3y - z)\bar{\jmath} + (4x + cy + 2z)\bar{k}$ is irrotational. Also find ϕ such that $\bar{A} = \nabla\phi$. 10M

15. A). Verify green's theorem in the plane for $\oint (x^2 - xy^3)dx + (y^2 - 2xy)dy$ where" is the square with vertices $(0,0), (2,0), (2,2)$ and $(0,2)$. 10M

OR

15. B). If $\bar{F} = (2x^2 - 3z)\boldsymbol{i} - 2xy\boldsymbol{j} - 4x\boldsymbol{k}$ then evaluate and $\int_v (\nabla X \bar{F})\, dv$ where 'v' is the closed region bounded by $x = 0, y = 0, z = 0, 2x + 2y + z = 4$. 10M

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